

HYDRAULIC MACHINES

Module 3

Module 3

Hydraulic Turbines & Pumps : Introduction, Classification, Construction details and working of Pelton, Francis and Kaplan turbines, Specific speed. Classification of water pumps, working of centrifugal pumps and reciprocating pumps (Theory of working principles only)

Power plants: Hydro-electric power plants, Thermal power plants, Nuclear power plants, Diesel power plants, Wind mills, solar energy (Working principles using schematic representations only)

- **Hydraulic machines** are defined as those machines which convert hydraulic energy in to mechanical energy or vice versa.
- Hydraulic Machines convert

Hydraulic energy  Mechanical energy eg : Turbine

mechanical energy  Hydraulic energy eg: Pumps

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- Mechanical energy developed by turbines is used to run electric generators coupled to the shaft of turbines
 - Hydro electric power is the most cheapest source of power generation

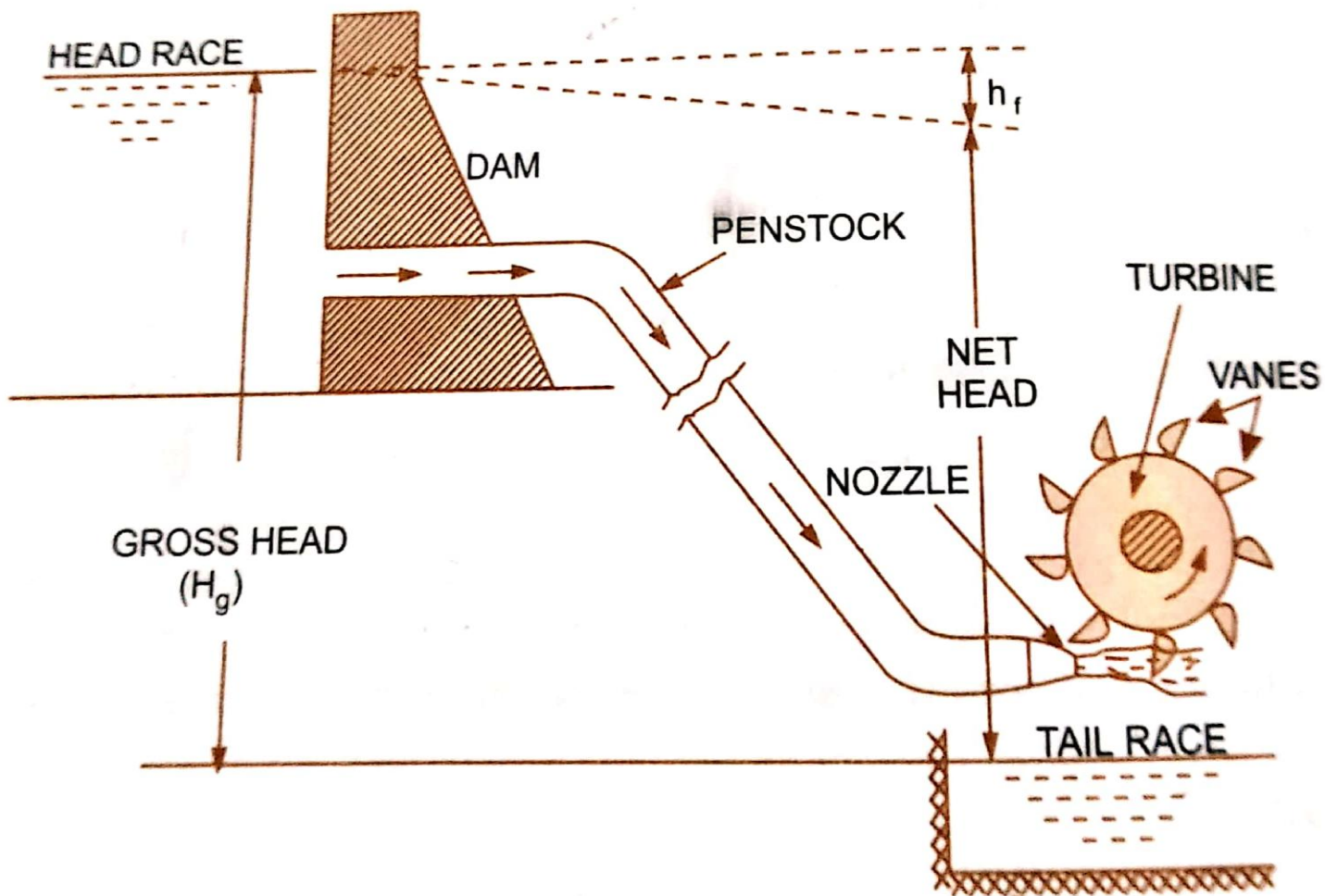
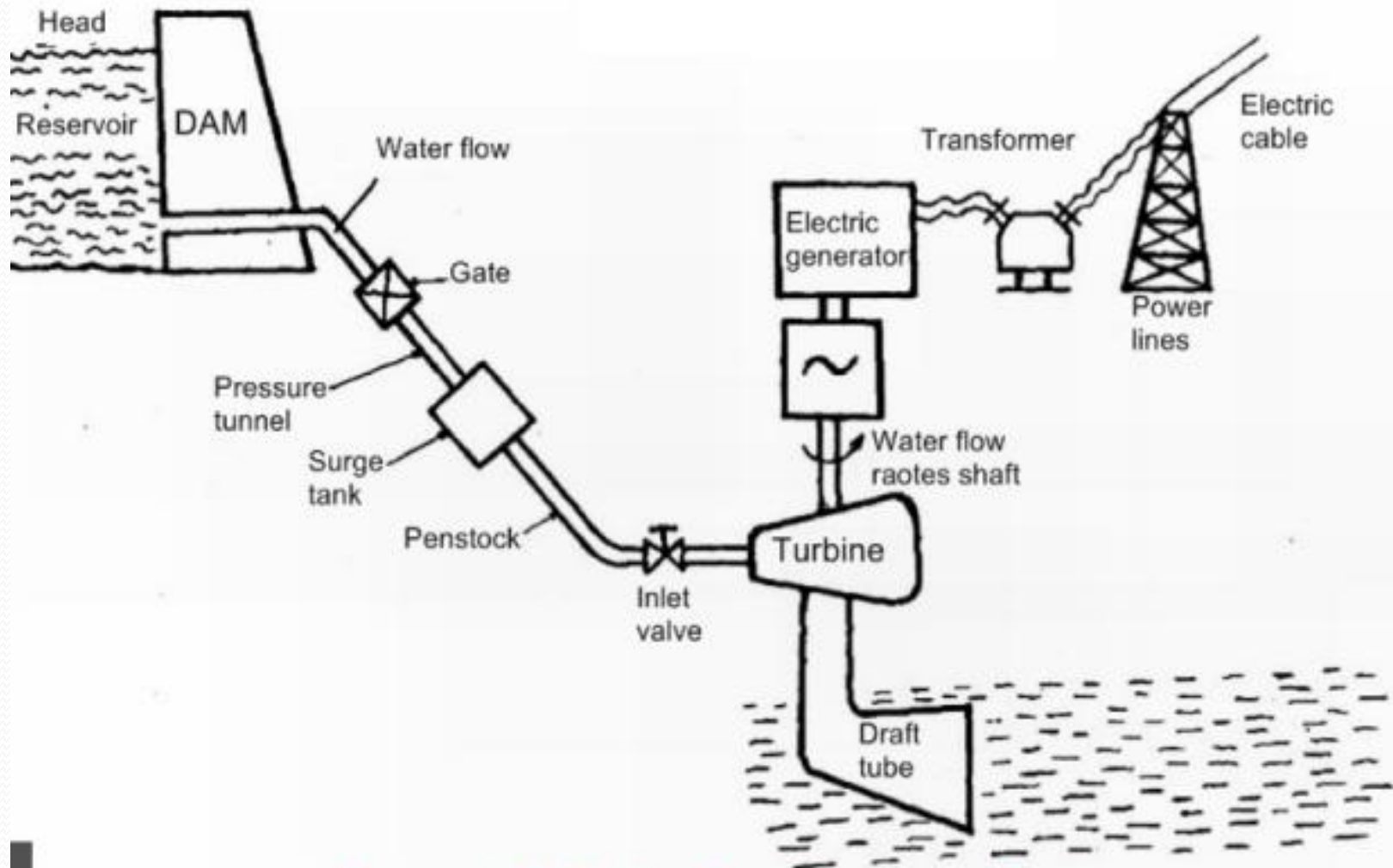


Fig. 18.1 Layout of a hydroelectric power plant.



Layout of hydro electric power plant

Classification of turbines

1. Based on type of energy at inlet

- a) **Impulse turbine**: Water possess only kinetic energy at the inlet of the turbine, eg. Pelton turbine
- b) **Reaction turbine**: Water possess both kinetic energy and pressure energy at the inlet, eg. Francis and Kaplan turbine

2. Based on head and discharge

- a) **High head turbine**: Head is more than 250m, low discharge, eg. Pelton turbine
 - b) **Medium head turbine**: 60m to 250m, medium discharge types, eg. Francis turbine
 - c) **Low head turbine**: Head will be below 60m, high discharge, eg. Kaplan turbine

3. Based on specific speed

1. Low specific speed (10 – 60) eg: Pelton
- 2 . Medium sp speed (60 – 300) eg: Francis
- 3 . High sp speed (300 – 1000) eg : Kaplan

4. Based on direction of flow through runner

1. Radial Flow turbine eg : Francis turbine
- 2 . Axial flow eg : Kaplan
- 3 . Mixed flow eg : new Francis
- 4 . Tangential flow eg : Pelton wheel

Impulse turbine

- the energy available at inlet of turbine, is only kinetic energy.
- Before reaching the turbine the fluid's Pressure head is changed to velocity head by flowing through nozzle.

Eg. Pelton wheel

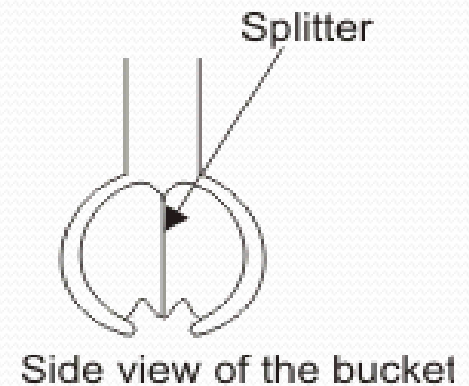
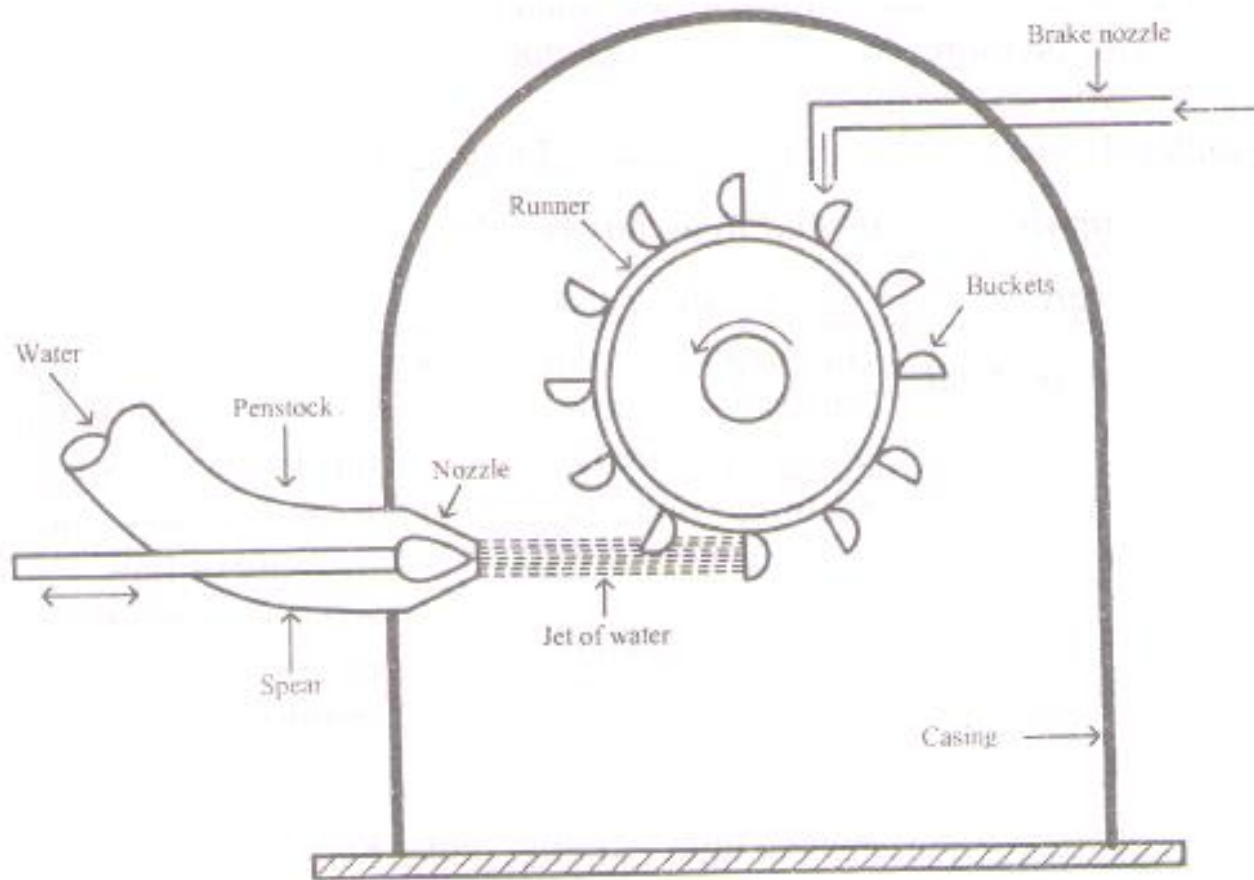
- Runner changes the direction of flow of a high velocity fluid jet.

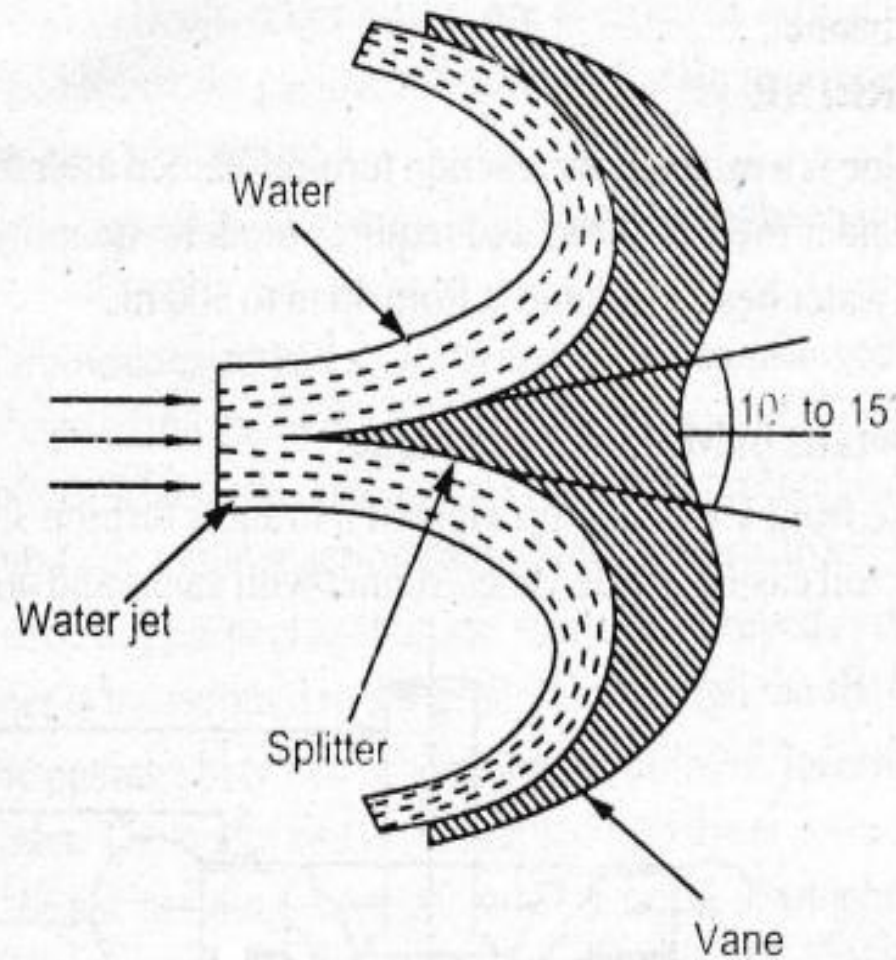


Reaction turbine

- Water at inlet possess KE as well as PE
- The pressure of the fluid reduces as it passes through the turbine runner blades, pressure energy changes to kinetic energy.
- This change in pressure energy is termed as reaction
- Requires water tight casing
- Eg: Francis Turbine, Kaplan Turbine etc.

Impulse Turbine - Pelton turbine

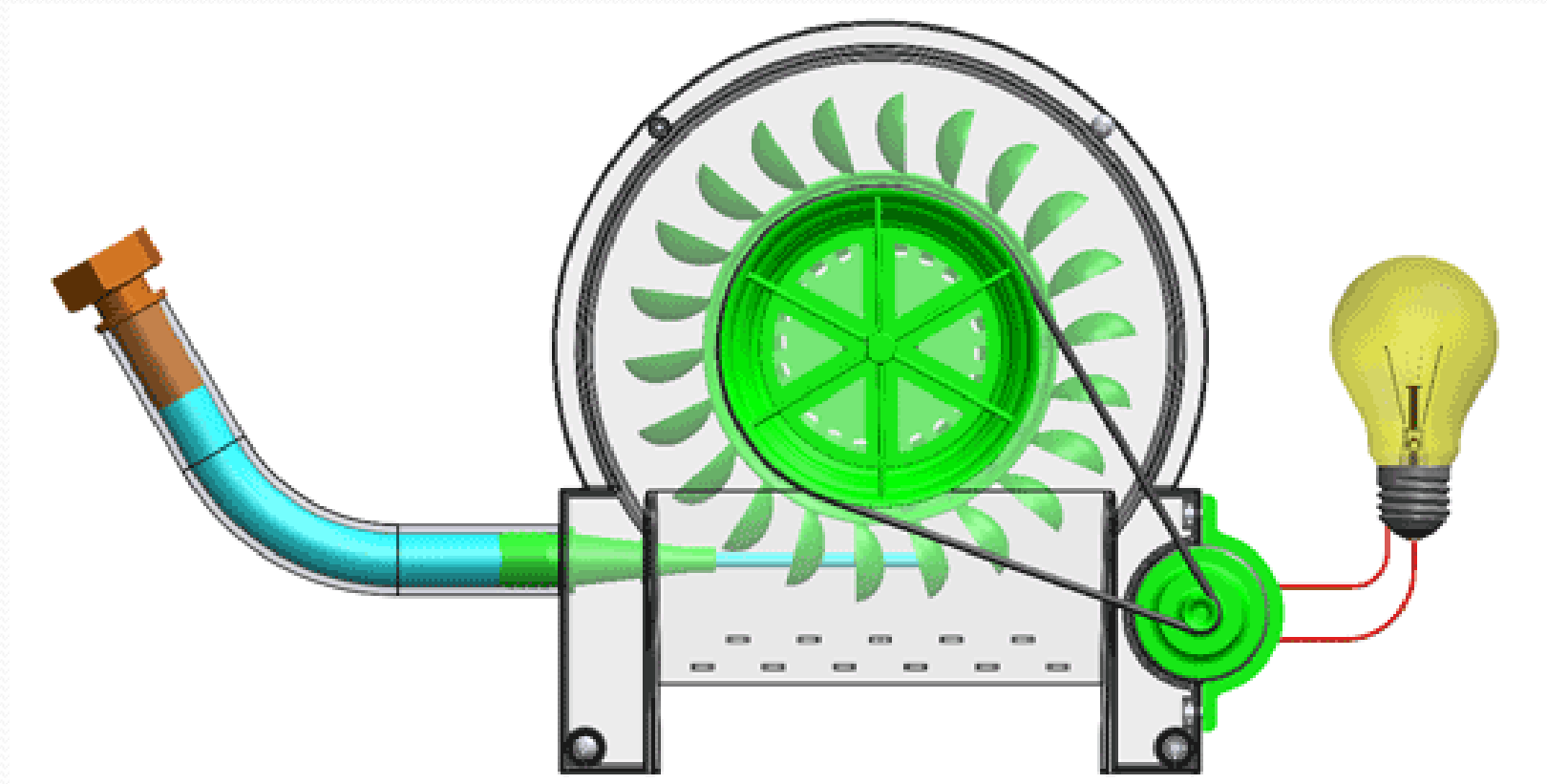




Shape of vane







Main Parts of a Pelton Turbine

- Nozzle and flow regulating arrangement
- Runner and buckets: Runner is a circular disc on the periphery of which a number of buckets are fixed.
- Casing: Prevent the splashing of water
- Breaking jet: Used to stop the runner

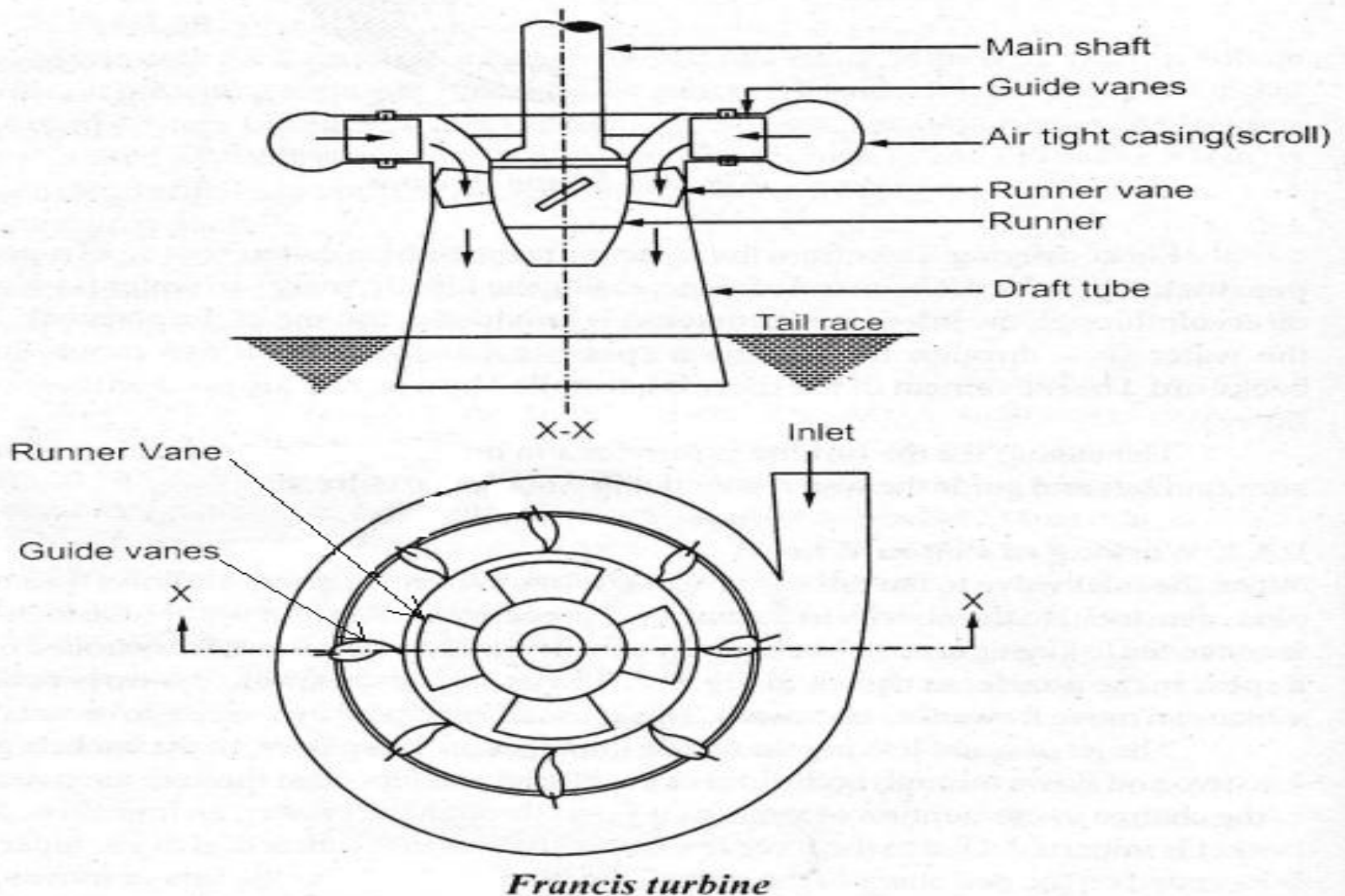
Working of Pelton wheel

- Nozzle directs the water against buckets mounted around the runner.
- When the water jet strikes the bucket, the water exerts impulsive force on the bucket and hence the runner revolves.
- The runner shaft is connected with the generator, thus the electricity is produced .

FRANCIS TURBINE

- Francis turbine is a radial flow or mixed flow reaction turbine
- It is well suited for medium head operations such as head ranging from about 60 to 250 meters.
- The major parts of Francis turbine are
 - ❖ Penstock,
 - ❖ Spiral casing,
 - ❖ Guide vanes,
 - ❖ Runner
 - ❖ Draft tube.

Francis Turbine

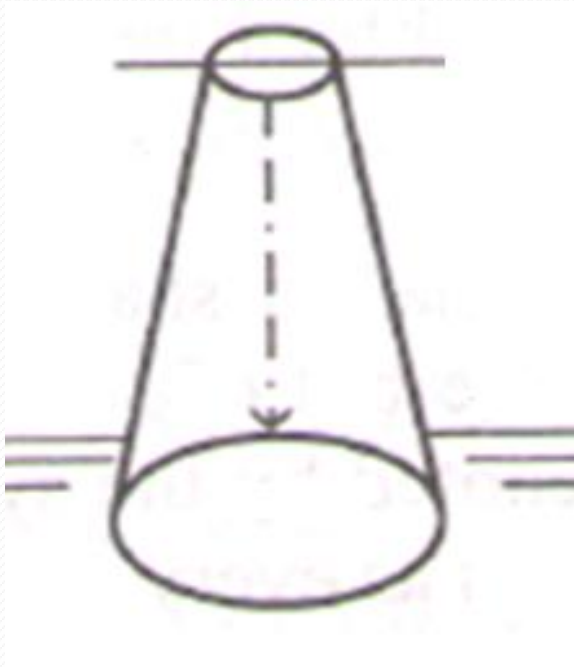


Main parts of Francis Turbine

- **Spiral casing**: Maintain a uniform velocity around the guide vanes
- **Guide vanes**: Around the circumference of the runner. It acts like a nozzle to increase the velocity of water. It also regulates the amount of water inlet to the turbine
- **Runner and runner vanes**: Runner vanes have aerofoil like structure. Pressure difference on the blades cause the rotation
- **Draft tube**: Water flows from runner outlet to tail race through draft tube

Draft Tube

It is a pipe of gradually increasing area which connects runner outlet to tailrace



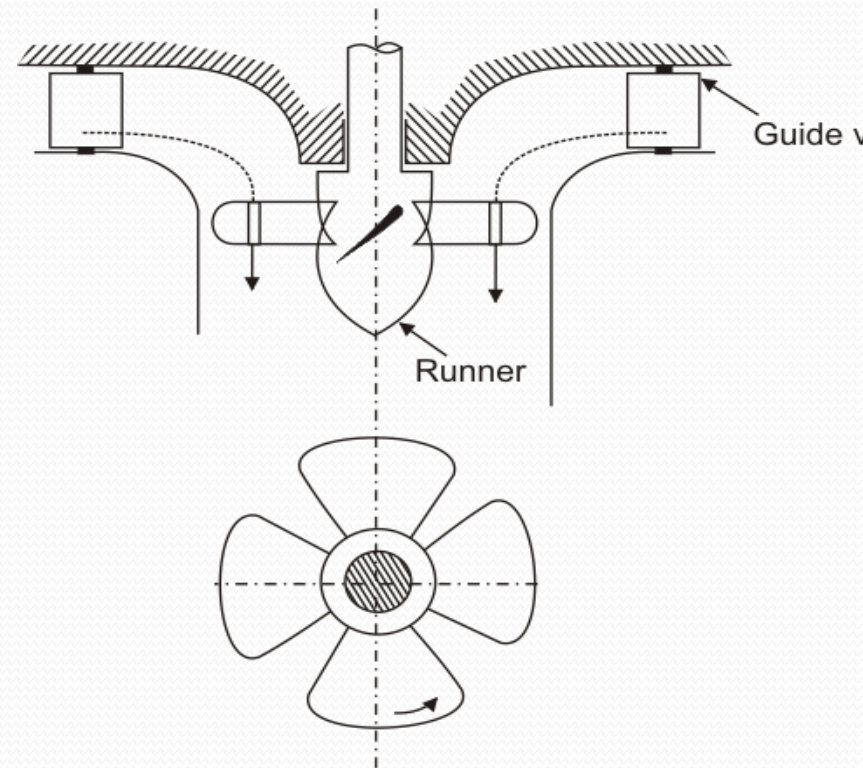
**Turbine exit.
Less area, Velocity more,
pressure less than atm**



**More area, Velocity less,
more pressure**

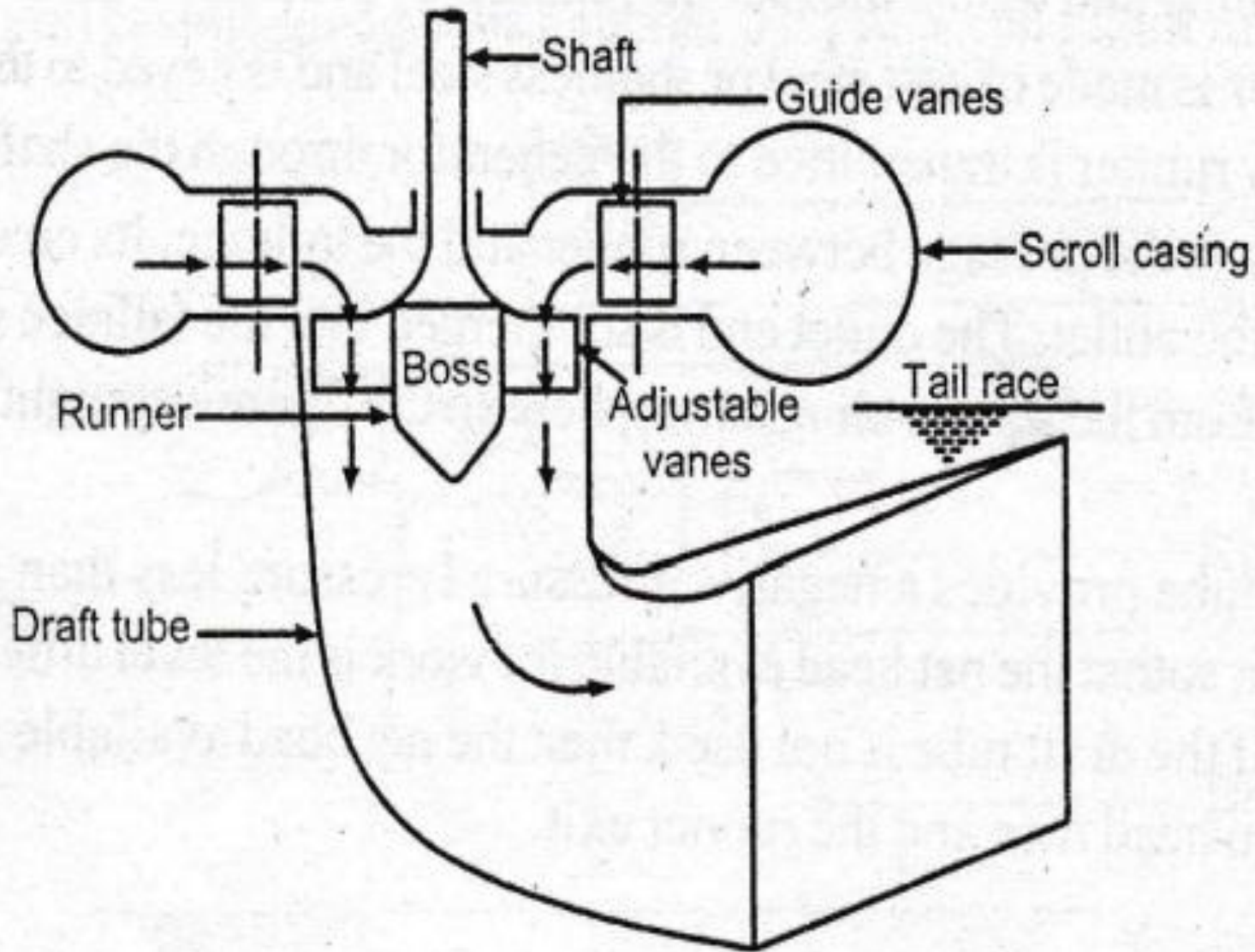
Kaplan turbine

- It is axial flow reaction turbine
- Low head reaction turbine
- Water enters to the runner axially and leaves axially
- Guide vanes and runner blades are adjustable



Main parts of a Kaplan Turbine

- Scroll casing
- **Guide vanes**: Used to turn the water through 90°
- **Hub and vanes**: Vanes are fixed to hub. Vanes are adjustable. Vanes are adjusted according to the flow rate. Vanes are aerofoil shaped profile
- Draft tube



Kaplan Turbine



Specific Speed (N_s)

- It is Speed of the turbine which is similar in shape, geometrical dimensions, blade angles, gate openings etc. with the actual turbine but of such a size that it will develop unit power working under unit head.
- It is denoted by the symbol N_s . The specific speed is used in comparing different types of turbines as every type has different specific speed.
- Specific Speed is give by the equation

$$N_s = \frac{N \sqrt{P}}{H^{5/4}}$$

P is Power Develo.
H is Head unde

working

N is Speed of actual turbine

Pump

- ***Device which converts mech energy to hydraulic energy.***
- ***Pump is a mechanical device which increases the pressure energy of a liquid.***
- ***Pump is generally used for raising liquid from low level to high level***

Classification

1. Positive displacement pumps

eg: Reciprocating pump, Gear pump

2. Roto dynamic pumps

eg: Centrifugal pump

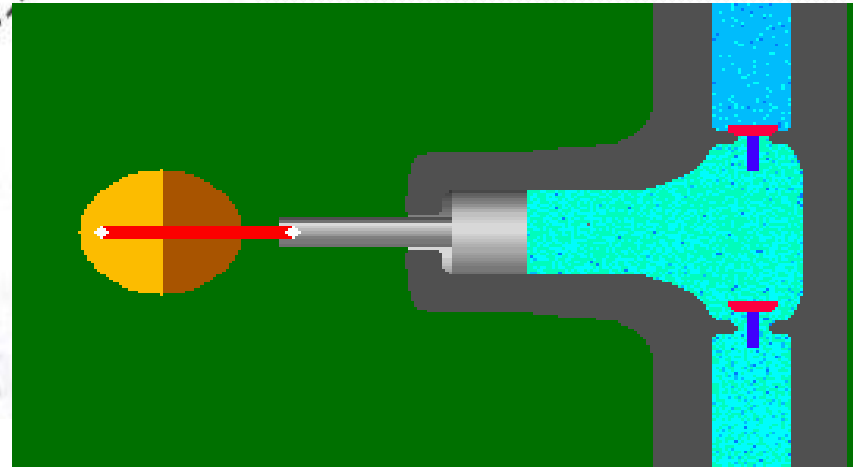
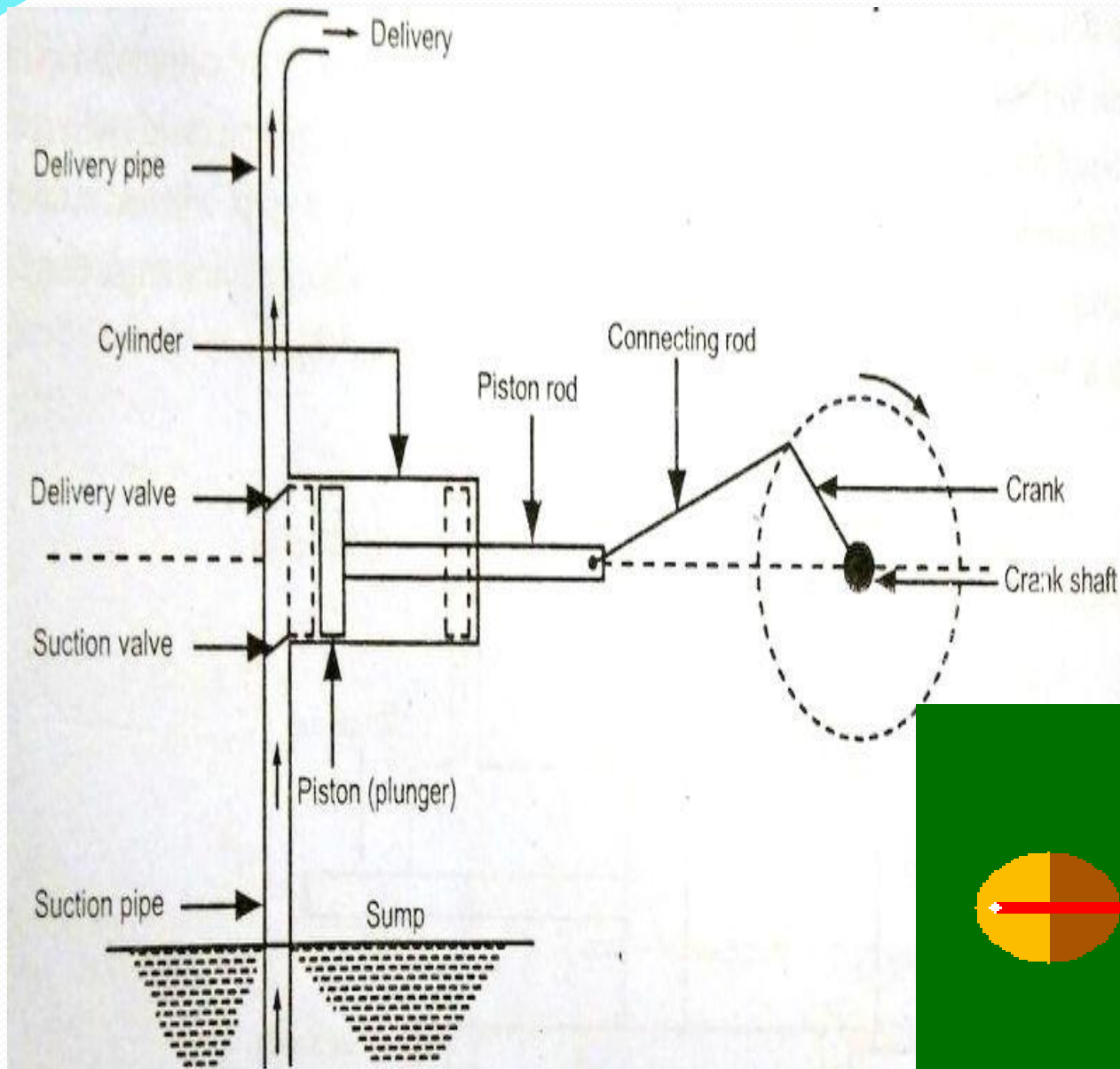
Positive displacement Pumps

- Reciprocating pump is a positive displacement pump.
- The positive displacement pump operates by alternating of filling a cavity and then displacing a given volume of liquid.
- It creates the lift and pressure by displacing the liquid using a moving mechanical element called plunger(piston) inside a cylinder.

Roto-dynamic pumps

- A centrifugal pump is a rotodynamic pump that uses a rotating impeller to create flow by the addition of energy to a fluid
- The roto-dynamic pump produces a head and a flow by increasing the velocity of the liquid through the machine with the help of a rotating vane impeller.

Reciprocating Pump



Main parts of a reciprocating pump

1)Cylinder with valves at inlet and delivery: Suction and delivery pipes with suction valve and delivery valve are connected to the cylinder. The suction and delivery valves are one-way valves or non-return valves, which allow the water to flow in one direction only

2)Plunger or piston: Piston reciprocates in the closely fitted cylinder

3)Connecting rod and crank mechanism: Crank and connecting rod mechanism is operated by a power source

4)Suction and delivery pipe with one way valve: One end of suction pipe remains dip in the liquid and other end attached to the inlet of the cylinder. One end of delivery pipe attached with delivery part and other end at discharge point

Working

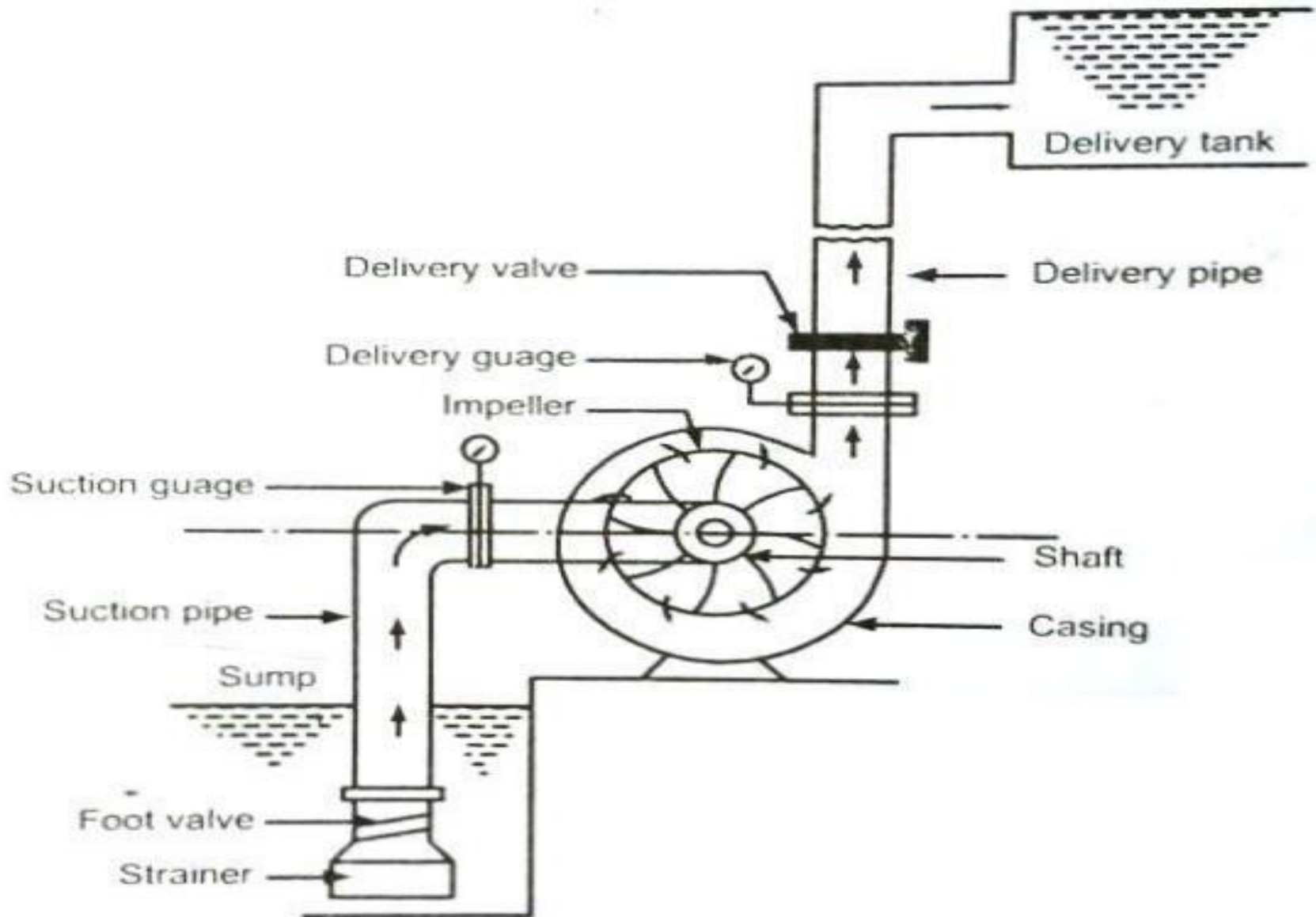
- Working similar to that of reciprocating engines
- Piston moves creates a vacuum inside the cylinder and atmospheric pressure forces the liquid up through the suction pipe into the cylinder.

Delivery valve will be closed during this stroke.

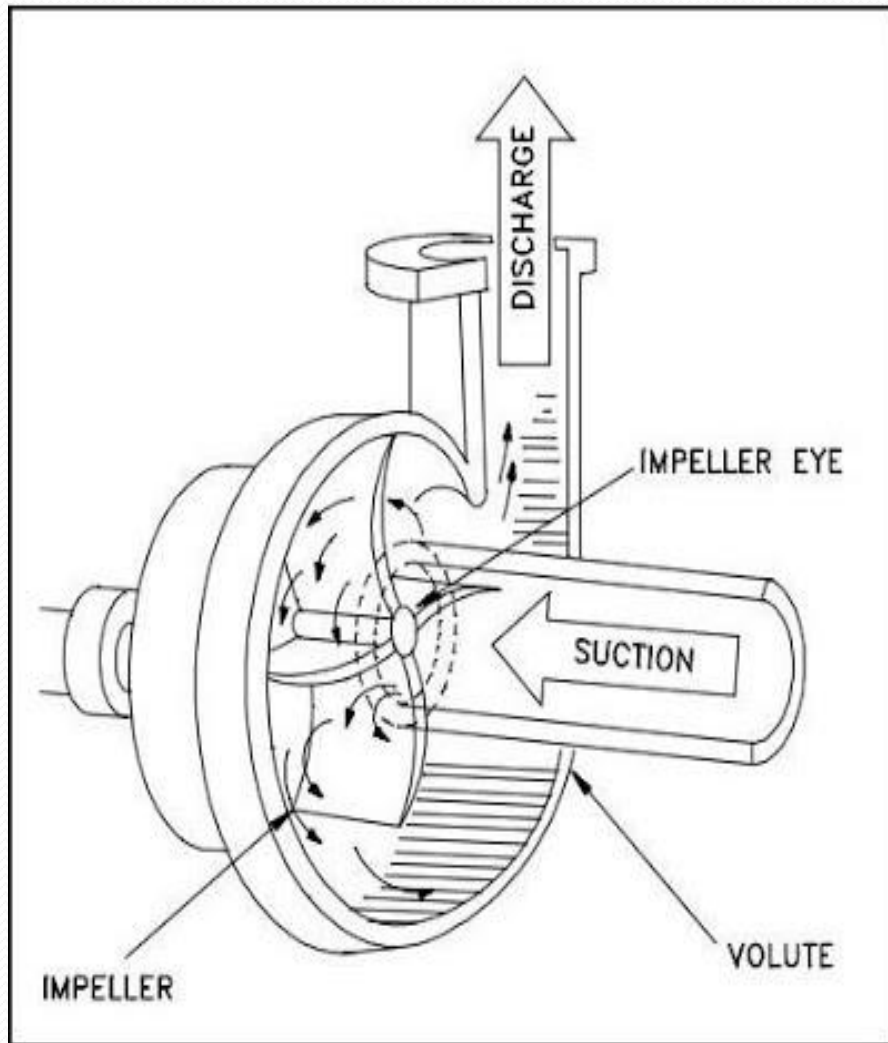
- During the return stroke, the pressure developed in the fluid opens the delivery valve, closes the inlet valve and pushes the fluid through the delivery valve.

- This pump is suitable for high heads and low discharge

CENTRIFUGAL PUMP



Centrifugal Pump



Parts of a centrifugal pump

1)Impeller: Rotating solid disc with curved blades. Impeller is mounted on a shaft connected to the shaft of an electric motor. As the impeller rotates, fluid is drawn into the impeller inlet(eye of pump) is accelerated as it is forced radially outwards

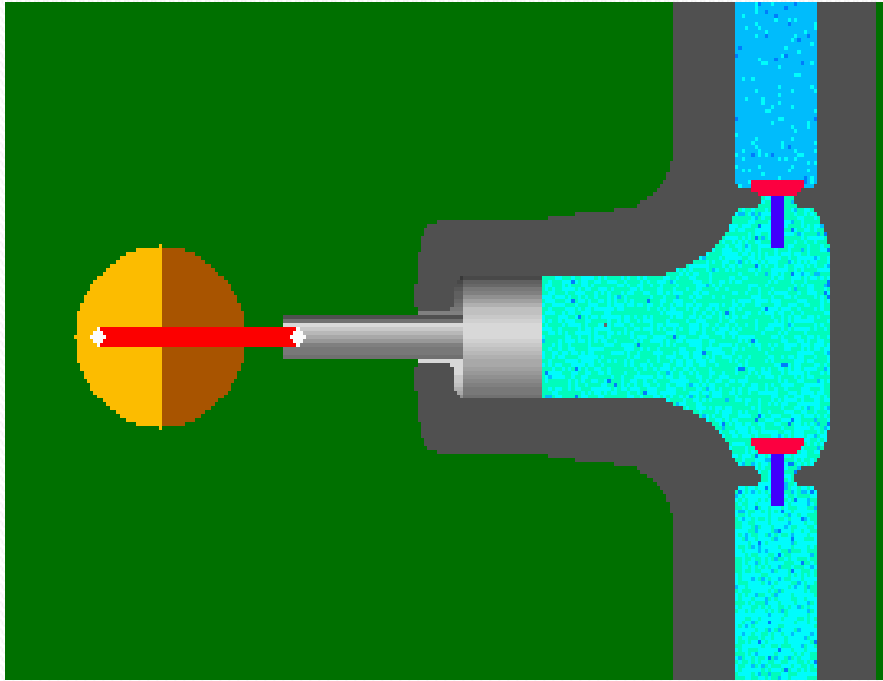
2)Casing: Air tight passage around the impeller

3)Suction pipe and delivery pipe: Pipe whose one end is connected to the inlet of pump and other end dipped in a sump is known as suction pipe.

Pipe whose one end is connected to the out let of the pump and the other end delivers the working fluid at a required height is known as delivery pipe.

Working

- Working fluid enters the pump at the centre of a rotating impeller.
- Impeller impart centrifugal force on the liquid entrapped in the impeller and throws the liquid towards the outer periphery of the impeller
- Outward movement of liquid in the impeller creates a partial vacuum near the eye of the impeller. Consequently, liquid from the sump is sucked in towards the impeller eye and enters through the inlet tip of impeller vanes. Thus, there is a continuous flow of liquid from the sump to the casing
- The liquid leaving the impeller vanes is at a higher pressure and velocity
- The velocity head is converted to pressure head in the casing



Reciprocating pump



centrifugal pump